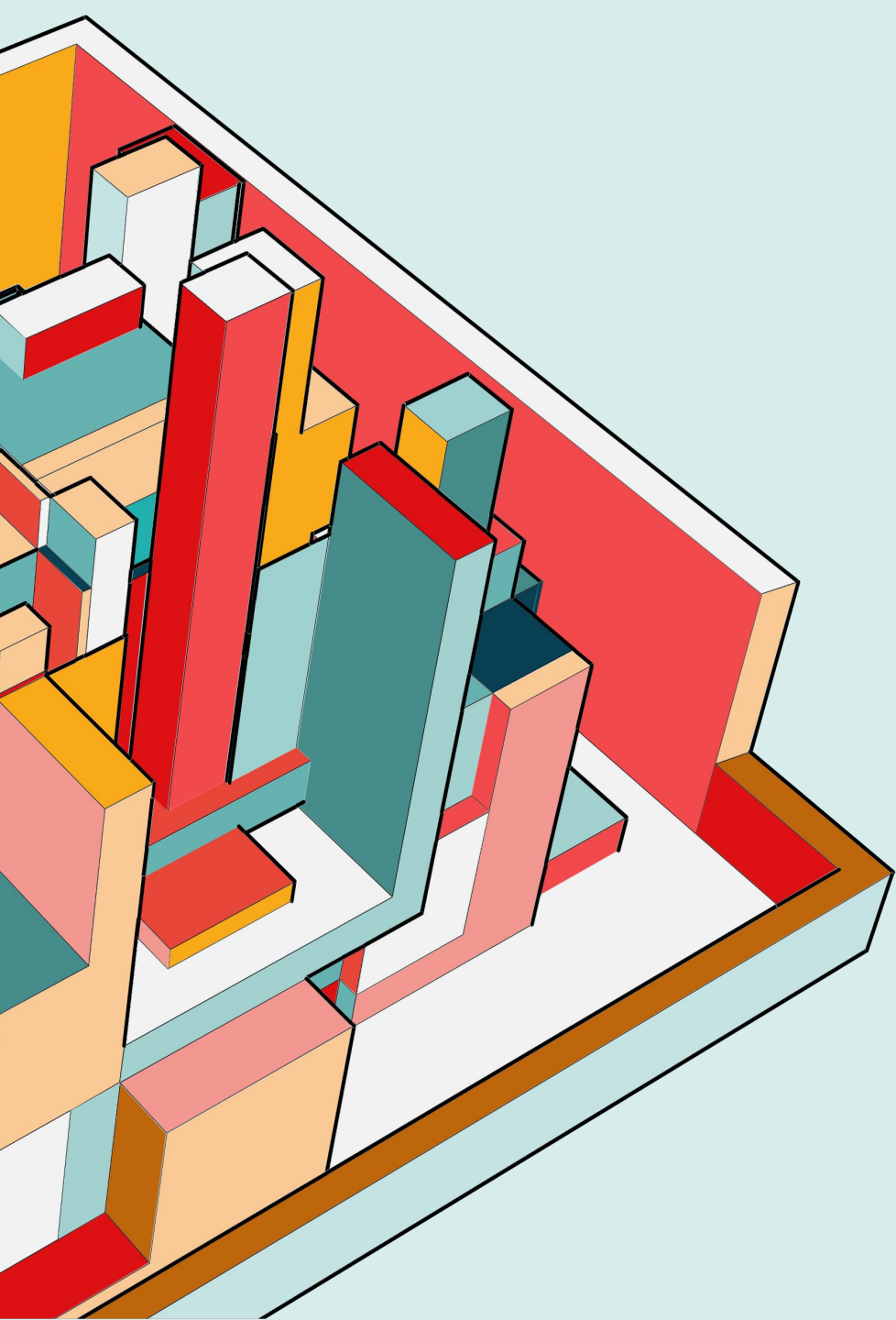




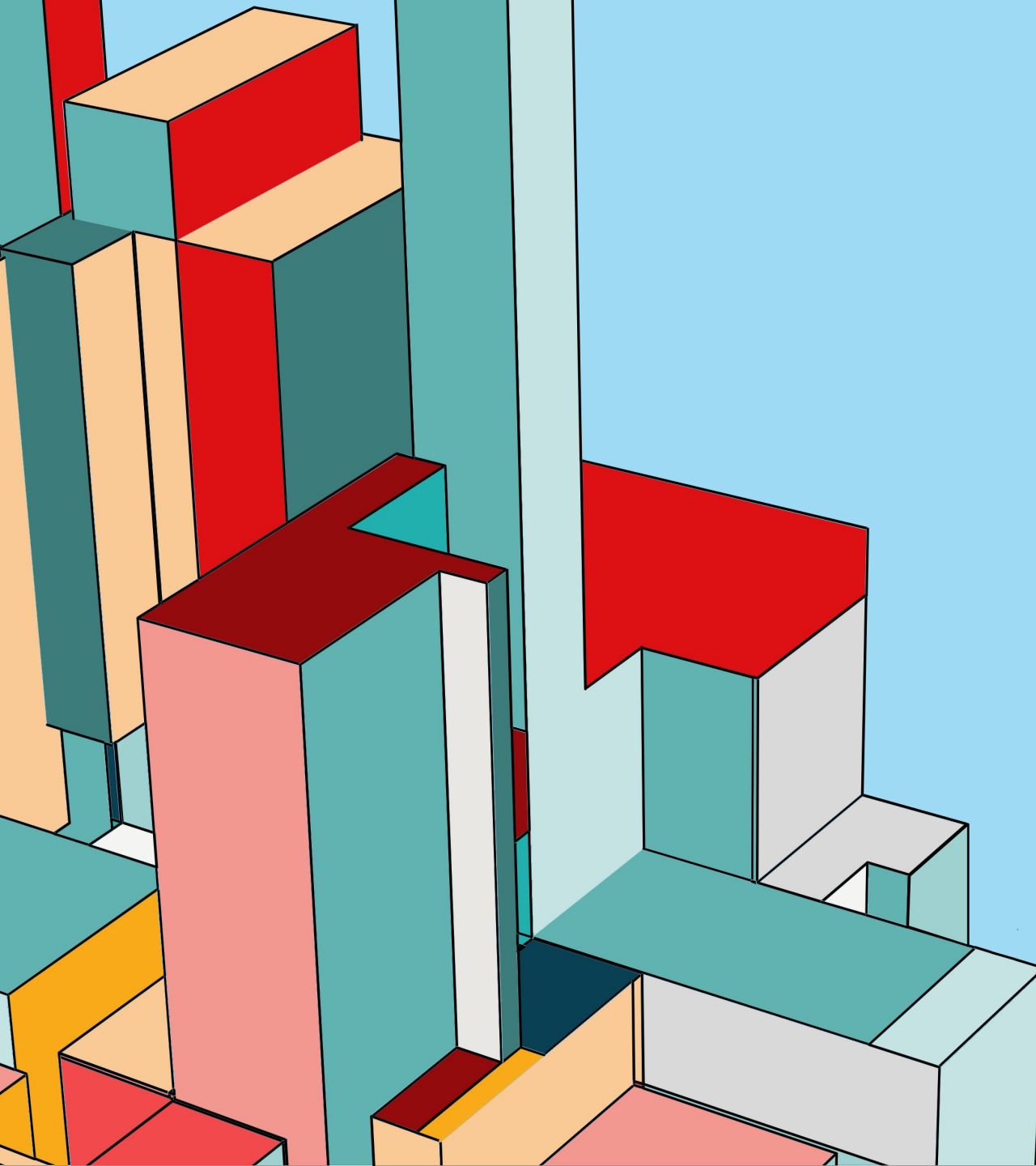
Methods 3 - Multilevel Statistical Modelling & Machine Learning

Classroom Instruction

WEEK 38

AGENDA

1. Covering last week + sorry
2. Today
 - a. Assignment_0
 - b. Explanations
3. Next week



Covering last week

A little bit on M3 and in particular readings

First part of M3 is centralized around one thing:

UNDERSTANDING MULTILEVEL MODELING

Just like M2 was all about OLS (hopefully still rings a bell). I will work with Lau to boil it down to this, revolving around 1-3 core concepts.

When reading you will see a lot of examples. They are nice for understanding, but rather skip them and prioritize understanding the core idea, e.g. how intercepts and slopes can vary across groups and why that is REALLY NICE for us modelling peeps :-)

So read with that in mind, since readings are LOVELY many pages. No irony detected. Second part of M3 is focusing on machine learning, which again will have 1-3 core concepts.

Sorry

Sorry - I did not know GitHub closed their classroom functionality.

All course material will be on Lau's dedicated repository (also assignment description)

Assignment will be handed in on Brightspace (feedback only for study groups handing in collectively, since individual feedback is not feasible).

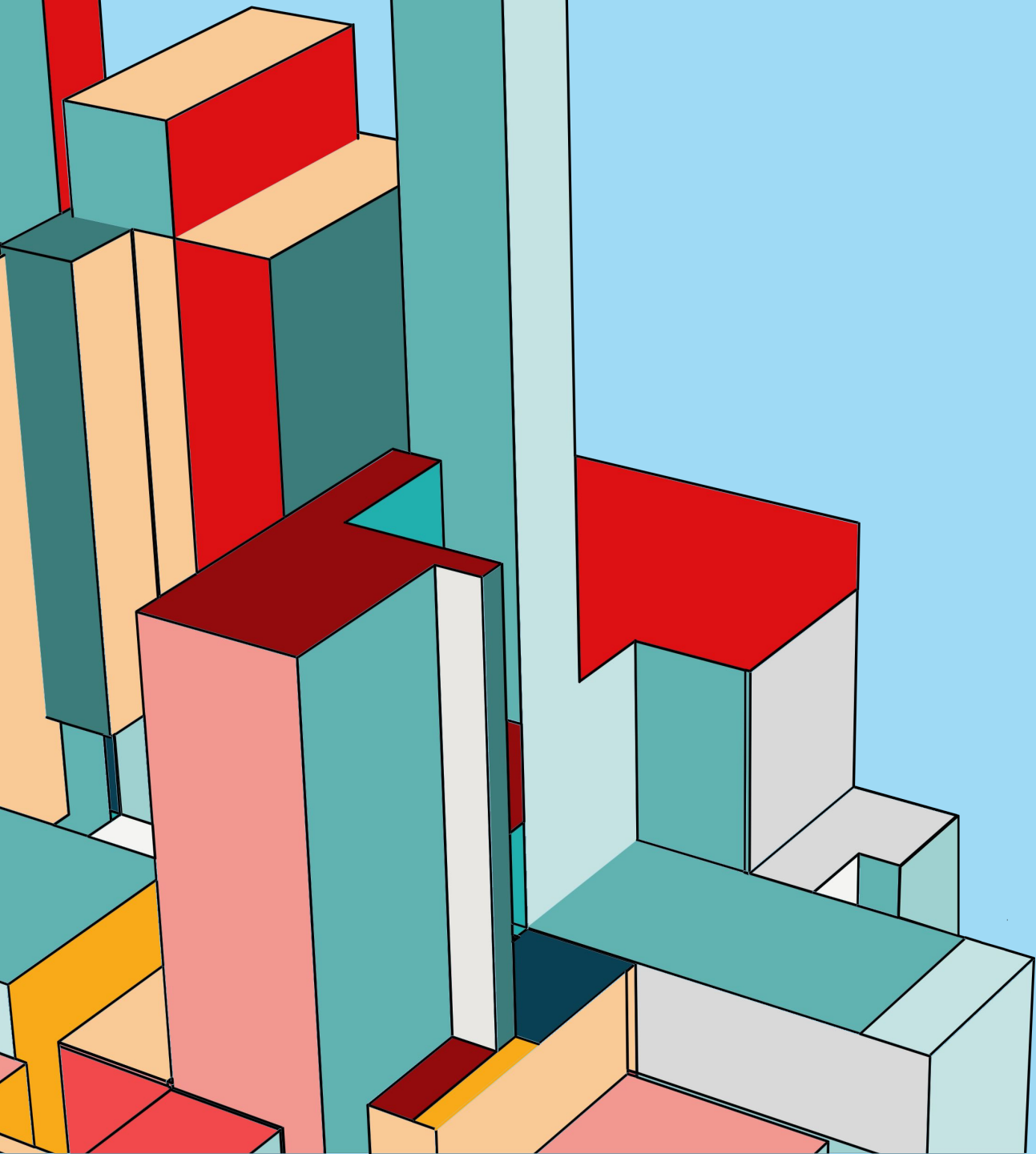
Also, I promised a full recap on last week's readings. Last week + this week readings is +200 pages (yay), but my brain can only *brain*, well, less than that. So whatever questions you have for the readings, I will be ready to discuss. I will occasionally bring up some core concepts, I find important.

What is M3 about?

Learning as much about multilevel modelling and machine learning as possible.

This does not mean learning as much possible about prompting a GAI.

Motivation not to use GAI: The bar is really low for passing M3 exam, and I believe you are ALL able to pass without use of GAI <3



Today

Get started on assignment_0

The screenshot shows the Canvas LMS interface. At the top, the course title 'Methods 3: Multilevel Statistical Modeling and Machine Learning (E26.147201U013.A)' is displayed. Below the title bar, there are navigation links: 'Course Home', 'Content', 'Activities', 'My Course', and 'Help'. The main content area is divided into three sections: 'Content Overview' showing a 'GitHub' link with a '0%' progress indicator, 'Calendar' showing the current date 'Thursday, 17 September 2026' and a list of 'Upcoming events' including 'SEP 24 11:59 PM Assignment 0 - Due', and 'Contacts' with a 'Setup' link. The 'Upcoming events' section is highlighted with a red box.

Read everything in here ----->

The screenshot shows the Canvas LMS assignment page for 'Assignment 0 - Due'. The due date is '24 September 2026 11:59 PM'. The assignment title is 'Methods 3: Multilevel Statistical Modeling and Machine Learning (E26.147201U013.A)'. The text says 'Here's the link for the assignment: [assignment_0](#)' and 'and the data that goes with it: [politeness data](#)'. Below this, there is a section titled 'Some remarks about I grade' with a list of requirements: 'don't show summaries of models', 'only show what I need to see', 'Remember to include the code that would give you your result', 'provide the pdf of the knitted Rmd and the Rmd', 'please don't answer questions in comments', 'Remember to use the natural units, e.g. for freq it is Hz', 'Plot meaningful x-axes and y-axes', 'for pdf's make sure that you stay within margins', and 'indicate clearly who wrote what - whenever author changes, use the following:'. Below the list, there is a code block for JavaScript:

```
1 summary(model) # no no!
```

. At the bottom, there is a section titled 'Detailed written feedback will be provided for Assignment 0 only, with the aim of aligning expectations of the instructor and students regarding how assignments will generally be read and graded. This has to be handed in as a study group assignment - otherwise you do NOT get feedback. Oral feedback will be provided on subsequent portfolio assignments during in-class assignment preparation sessions, where students will have ample opportunity to ask the instructor questions on their assignments-in-progress.'

Explaining core concepts

Notation clarification

$$Y = \overset{\text{Level 1}}{\boxed{X\beta}} + \overset{\text{Level 2}}{\boxed{Zb}} + \epsilon$$

Fixed effects Random effects

$$b \sim N(0, \hat{\Sigma})$$


Difference between level 1 and level 2:
We model the values in the b-vector to be normally distributed (because they are related).

This is a matrix, not a sum!

Explaining core concepts

Understanding this:

$$y = XB + Zu + e$$

Is THE ONE main goal in M3 (first part).

You remember that OLS was that in M2:

$$y = XB + e$$

where

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

The picture is from when Lau taught me M2 stuff. Let it trigger happy memories from our M2 last semester.

The whiteboard contains handwritten mathematical derivations for OLS. At the top, there are two sets of dimensions: $3, 6, 6, 3$ and $3, 6, 6, 1$. The main derivation is for the OLS estimator $\hat{\beta}$:

$$\hat{\beta} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}^{-1} @ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} @ \begin{bmatrix} 10 \\ 4 \\ 15 \\ -2 \end{bmatrix}$$

Below this, there are intermediate steps:

$$Z = X^T X$$
$$Z^{-1} Z = I$$
$$n = \frac{1}{2} = 1$$

Then, the matrix Z is calculated as:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}^{-1} @ \begin{bmatrix} 5 \\ 14 \\ 13 \end{bmatrix}$$

The dimensions $3, 3$ and $3, 1$ are noted below the matrices. Finally, the result is:

$$\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} @ \begin{bmatrix} 5 \\ 14 \\ 13 \end{bmatrix} = \begin{bmatrix} 2,5 \\ 7 \\ 6,5 \end{bmatrix}$$

Explaining core concepts

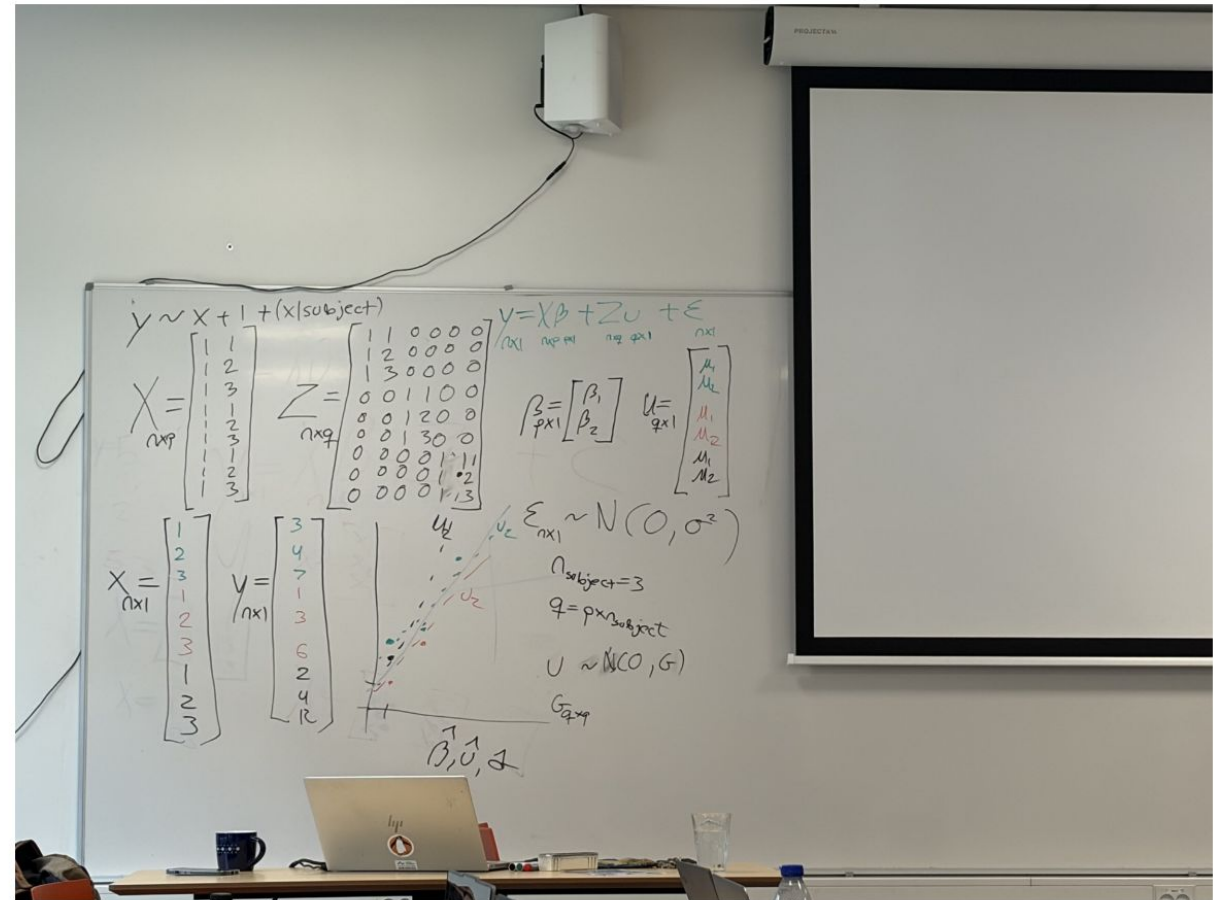
Understanding this:

$$y = XB + Zu + e$$

Is THE ONE core concept.

This is a picture of Lau's class last week.

Let's engage it.



$x_{n \times 1}$ is the independent variable from three different subjects, colour coded.

$y_{n \times 1}$ is the dependent variable, measured from the same subjects, colour coded.

$X_{n \times p}$ is the design matrix for the general effects across participants (first level effects) (intercept and slope)

$Z_{q \times p}$ is the design matrix for the effects for each participant (second level effects) (intercept and slope for each subject)

$\beta_{p \times 1}$ are the coefficients for the two parameters of X

$u_{q \times 1}$ are the coefficients for the six parameters of Z ($q = n_{subjects} \times p$)

$\epsilon_{n \times 1}$ are the residuals, assumed to normally distributed with a mean of 0 and a variance of σ^2

u is normally distributed with a mean of 0 and a variance-covariance structure $G_{q \times q}$

Explaining core concepts

Notation clarification

$$Y = \overset{\text{Level 1}}{\boxed{X\beta}} + \overset{\text{Level 2}}{\boxed{Zb}} + \epsilon$$

Fixed effects Random effects

$$b \sim N(0, \hat{\Sigma})$$

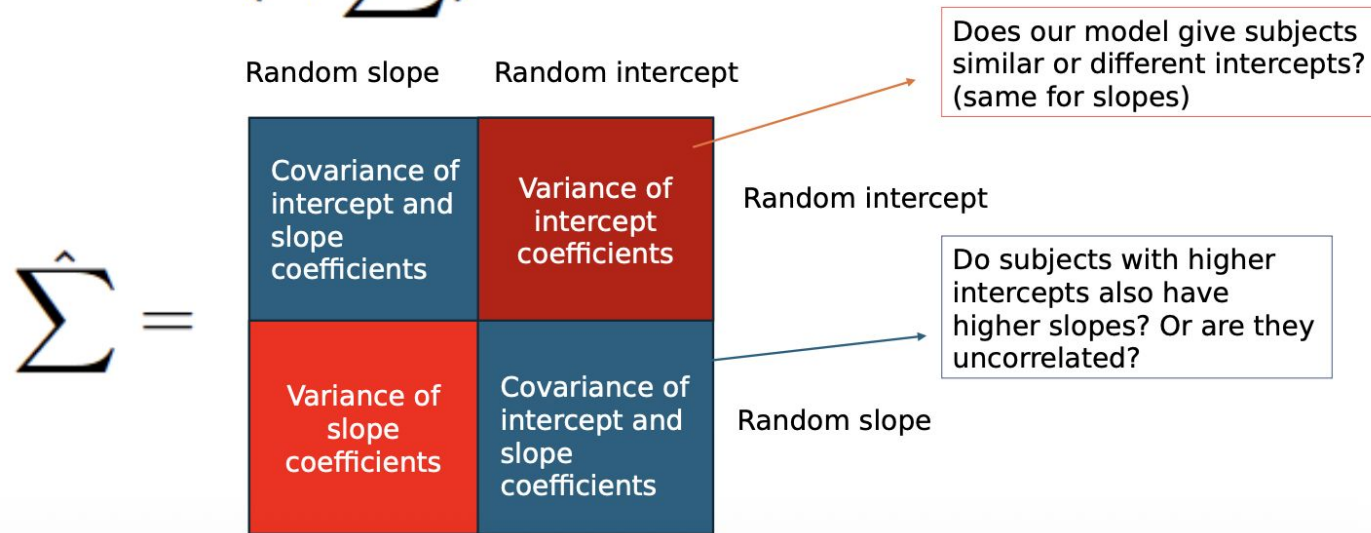

Difference between level 1 and level 2:
We model the values in the b-vector to be normally distributed (because they are related).

This is a matrix, not a sum!

What is this weird E? It's a matrix - not a sum!

Notation clarification

$$b \sim N(0, \hat{\Sigma})$$



$$s^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

$$Cov(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Annotations for the covariance formula:

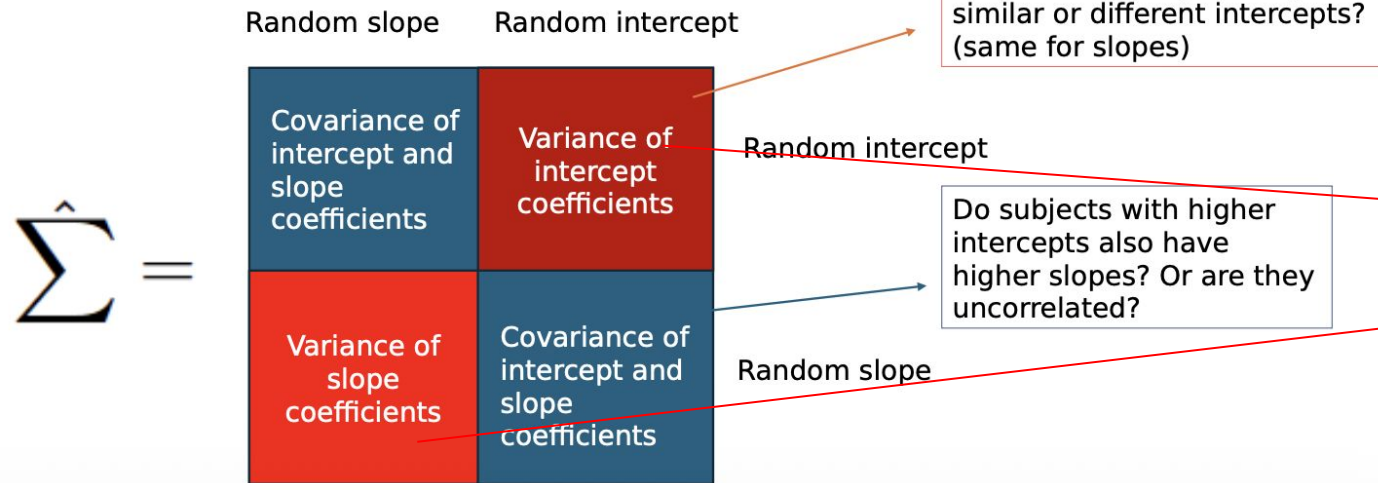
- n : total count of sample values
- x_i : single observed value of dependent variable
- $\bar{x}$: mean of all values of independent variable
- y_i : single observed value of independent variable
- $\bar{y}$: mean of all values of independent variable
- $n - 1$: population count minus one (Bessel's Correction)

α

What is this weird E? It's a matrix - not a sum!

Notation clarification

$$b \sim N(0, \hat{\Sigma})$$



```
# wk38
```{r}

data("sleepstudy")
df <- sleepstudy

model1 <- lmer(Reaction ~ Days | Subject , data = df)
VarCorr(model1)
```
```

| Groups | Name | Std.Dev. | Corr |
|----------|-------------|----------|--------|
| Subject | (Intercept) | 25.527 | |
| | Days | 11.927 | -0.182 |
| Residual | | 25.592 | |

How to use the lme4 package in R

General how-to in R

Random intercept:

```
Model <- lmer(reaction ~ sleep_deprivation + (1 |Subject), data=df)
```

Random slope only model:

```
Model <- lmer(reaction ~ sleep_deprivation + (0+sleep_deprivation |Subject),  
data=df)
```

Random slope and Intercept:

```
Model <- lmer(reaction ~ sleep_deprivation + (sleep_deprivation|Subject ), data=df)
```

Using lmer() automatically uses pooling. For non-pooled intercept use:

```
lm(reaction ~ sleep deprivation + Subject, data = df)
```

For non-pooled intercept and slope use:

```
lm(reaction ~ sleep deprivation * Subject, data = df)
```

The scary thing from lecture

Partial pooling (shrinkage) of group coefficients α_j

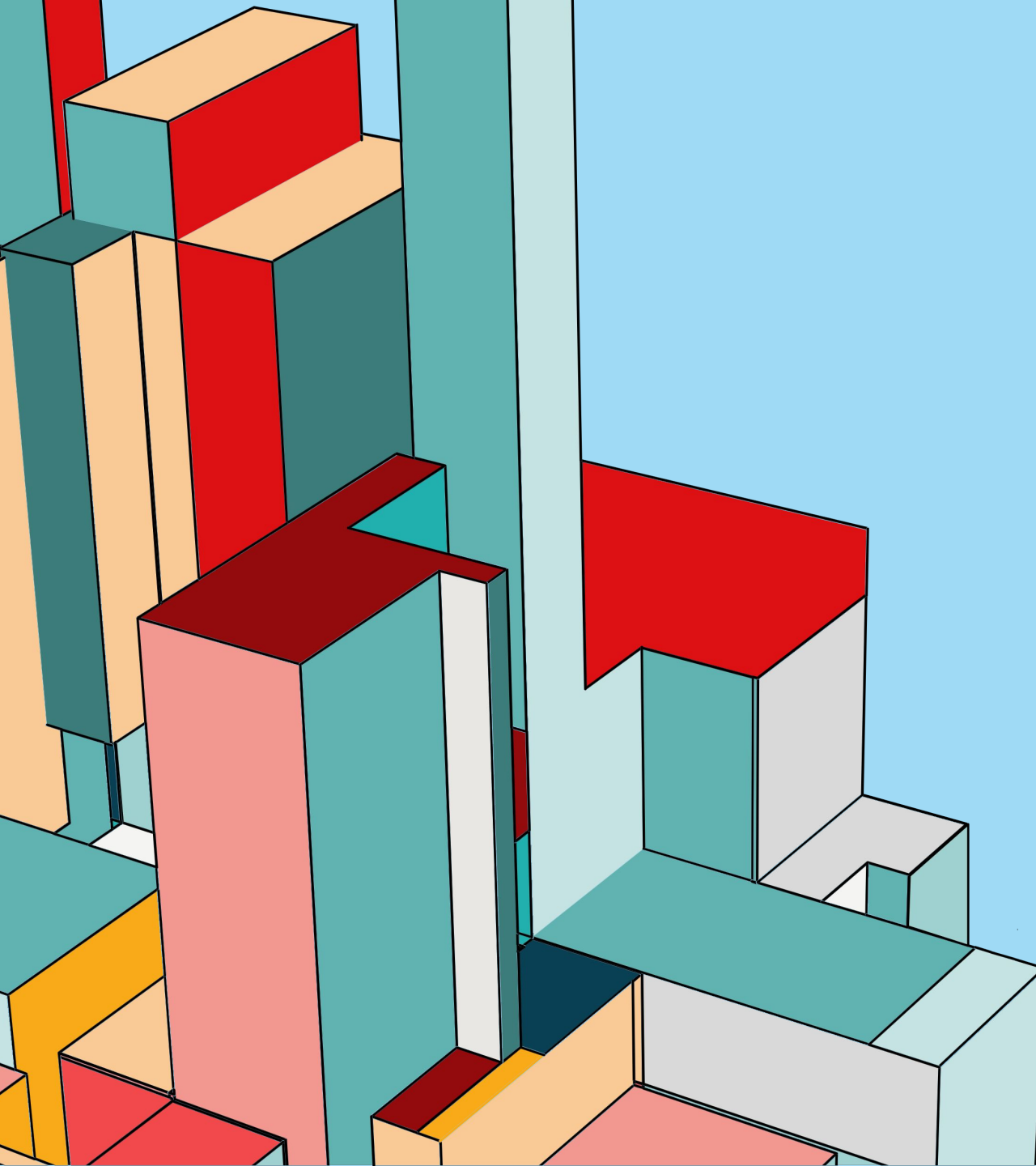
Multilevel modeling partially pools the group-level parameters α_j toward their mean level, μ_α . There is more pooling when the group-level standard deviation σ_α is small, and more smoothing for groups with fewer observations. Generalizing (12.1), the multilevel-modeling estimate of α_j can be expressed as a weighted average of the no-pooling estimate for its group ($\bar{y}_j - \beta\bar{x}_j$) and the mean, μ_α :

$$\text{estimate of } \alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} (\bar{y}_j - \beta\bar{x}_j) + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} \mu_\alpha. \quad (12.4)$$

When actually fitting multilevel models, we do not actually use this formula; rather, we fit models using `lmer()` or `Bugs`, which automatically perform the calculations, using formulas such as (12.4) internally. Chapter 19 provides more detail on the algorithms used to fit these models.

No more chit chatting from me.

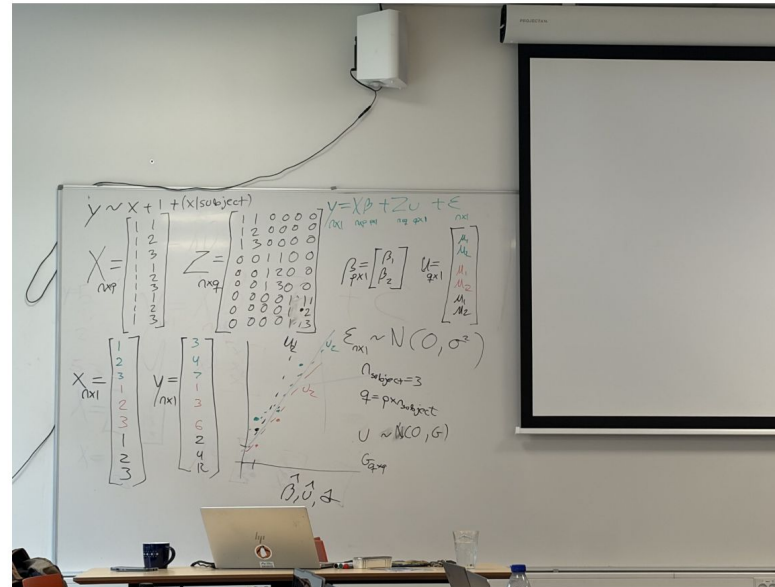
Let's hit it with assignment_0



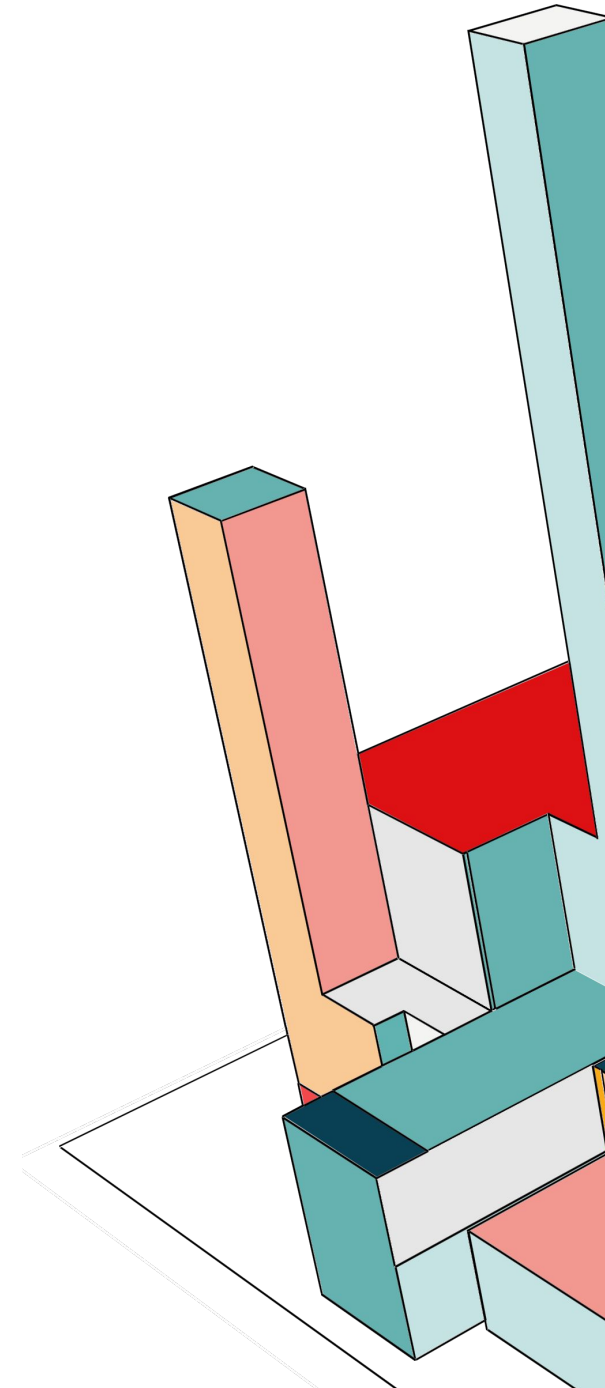
For Next Week

FOR NEXT WEEK

It's all about MULTILEVEL modelling. Keep that in mind when reading :-)



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SEE YOU NEXT WEEK!
insert silly picture

